Proficient In Revit Architecture (80 Hours)

S.No	Session	Topics	Type	Duration (Hours)
1	Session 1	**Introduction** About BIM Introduction to Autodesk Revit Architecture **Revit File Types** Understanding Project Project Templates Family Templates **Understanding User Interface** Home Tab (Ctrl + D) Quick Access Toolbar. Title bar Info Centre File menu Tabs (Architecture, Structure…) Panel (Build, Circulation, Room & Area…) Options bar Properties Palette (PP) Type selector Property filter Edit type properties Instant properties Project Browser window View control options Status bar Working space View cube **Building Elements** Defining Building elements Building Element Types **Starting a New Project** Starting a project using Template Using No Template and Saving Project Files	Theory/Demo	2.0 Hrs
2	Session 2	Exercises included for Practice	Practical	2.0 Hrs
3	Session 3 & 4	**Drawing Aids** Alignment Lines Snaps Project Units **Levels** Adding Levels Modifying Levels Creating Level Head Constrain Levels **Grids** Adding Grids Editing Grid Bubbles **Walls** Creating and Modifying Walls Create New Wall type Location Line Wall joins Add Partition Wall Wall layer wrapping Understanding Stacked Wall Creating Stacked Wall Guidelines for Creating Stacked Walls Vertically compound Wall Wall shapes and Openings Wall sweeps and Reveals **Construction Modelling Tools** Understanding Parts Divide Parts Dividing Parts by Sketching Dividing Parts by Reference Understanding Assemblies Creating Assemblies	Theory/Demo	4.0Hrs

		Creating Assembly View and Sheets		
		Placing anassembly on Sheet		
4	Session 5	Exercises included for Practice	Practical	2.0 Hrs
5	Session 6	Exercises included for Practice	Practical	2.0 Hrs
6	Session 7 & 8	**Modify Tools**	Theory/Demo	4.0Hrs
		Move		
		Move by Dragging		
		Moving End Joined Element,		
		Moves with Nearby Elements		
		Restriction toMove with Nearby Elements		
		Creating copies using Copy tool		
		Create Similar		
		Rotate		
		Mirror		
		Array		
		Linear Array		
		Radial Array		
		Scale		
		Resize the elements using scale tool		
		scaling numerically		
		resizing elements with shape handles		
		resizingusing formulas		
		Split Element		
		Understanding split element tool		
		Split with gap		
		Joining walls that are split with a gap between them		
		unjoining walls created using split with gap		
		Trim		
		Trim or extend to corner		
		Trim or extend single element		
		Trim or extend multiple elements		
		Align		
		Offset		
		Pin & Unpin Elements		
		Clipboard		
		Doors & Windows		
		Understanding the basics of Doors and Windows		
		Adding doors and windows		
		Moving Doors and windows to a different wall		
		Tape Measure tool		
		Keyboard Shortcuts		
		Adding Keyboard Shortcuts		
		Guidelines for adding keyboard shortcuts		
		Removing shortcuts		
		Import keyboard shortcuts & Export keyboard shortcuts.		
7	Session 9	Exercises included for Practice	Practical	2.0 Hrs
8	Session 10	Exercises included for Practice	Practical	2.0 Hrs
9	Session 11 & 12	**Floor**	Theory/Demo	4.0Hrs
		Adding Floor		
		Creating Sloped Floor using Slope Arrow		
		Creating Sloped Floor using Parallel Sketch Lines		
		Floor Slab Edge		
		Ceiling		
		Creating ceiling using automatic ceiling tool		
		Components		
		Placing different types of components		
		Working with Modern Medium Library		
		Roof		
		Adding Roof by Footprint		
		Adding Roof by Extrusion		
		Adding Sloped Glazing		
		Modifying Roof		
		Shape editing for Roofs and Floors		
		Roof Soffit, Fascia & Gutter		
		Openings		
		Opening on face and Vertical Opening		
		Wall opening,		

		Shaft opening		
		Dormer Opening		
10	Session 13	Exercises included for Practice	Practical	2.0 Hrs
11	Session 14	Exercises included for Practice	Practical	2.0 Hrs
12	Session 15	**Dimensions**	Theory/Demo	2.0 Hrs
		Temporary Dimensions and Permanent Dimensions		
		Spot Dimensions		
		Modifying Dimensions and Constraints		
		Managing Views		
		Creating a Plan View		
		View range		
		Plan region		
		Elevation View		
		Section View		
		3D Views		
		Cropping View		
		Visibility or Graphics Display in Project		
		View Templates		
		Duplicate Views		
		Controlling the visibility and graphicsdisplay using Filter		
		Visual Styles		
		Graphic Display Options		
13	Session 16	Exercises included for Practice	Practical	2.0 Hrs
14	Session 17 & 18	**Curtain Wall**	Theory/Demo	4.0Hrs
		Understanding the Curtain Wall		
		Creating Curtain Walls		
		Adding Curtain Grids		
		Curtain Grid snapping		
		Excluding grid from curtain panels		
		Mullions		
		Mullion Profiles		
		Controlling Mullion Join		
		Corner Mullions		
		Quad Corner Mullion		
		Trapezoid Corner Mullion		
		Adding Curtain Door to Panel		
		Embedded Walls		
		Understanding Embedded Walls		
		Embedding wall into Another Wall		
		Stairs		
		Creating Stair by component		
		Creating a Run Component		
		To Create Runs connected by Landings		
		Creating support component		
		creating landing component by picking two runs		
		Create Customized Stair by Sketch method		
		Modifying Stair sketch using direct manipulation controls		
		Customizing stair documentation		
		Modifying Cut mark		
		Annotating Stair Path		
		Tagging Stair Components		
		Stair calculator		
		Creation of Stair using Stair Calculator		
		Modify stair railings		
		Extent the stair railings		
		Ramp & Railings		
		Understanding Ramps and Adding Ramps		
		Add railings		
		Add railings by sketching		
		Modify railing		
		Modifying Railing structure		
		Modifying Rail Joins		
		Modifying Rail Height and Slope		
		Controlling the Placements of Balusters and Posts		
		Modifying continuous rail		
		Defining continues Rail Extensions		
		Modifying anExtension		

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		Modifying Handrail and Support		
15	Session 19	Exercises included for Practice	Practical	2.0 Hrs
16	Session 20	Exercises included for Practice	Practical	2.0 Hrs
17	Session 21 & 22	**Massing** Understanding the Mass Objects Conceptual Design Environment Understanding In-Place Mass **Forms** Understanding Forms Types of Forms Surface Forms Extrusions, Lofts, Revolve, Sweep, Swept Blend Modifying Forms Editing Profile Add Edge, Dissolving Elements, Locking Profiles, Rehosting Forms, DivideSurface Using Cut and Join Geometry Tools, Joining Geometry **Rationalizing Surface** Dividing Surface with UV Grids Patterning Surface Editing Pattern Surface ModifyingPattern Border Tiling Pattern Component Families Surface Pattern Representation Splitting Faces Applying and removing a paint using Paint tool Placing Mass Instances from Mass Family andFace based Modeling **Conceptual Energy Analysis** Energy Settings Enabling Energy Models Run Energy Simulation Results & Compare Customizing Form Rationalization and Controlling Visibility of Mass Structure	Theory/Demo	4.0Hrs
18	Session 23	Exercises included for Practice	Practical	2.0 Hrs
19	Session 24	Exercises included for Practice	Practical	2.0 Hrs
20	Session 25	**Documentation** **Text** Adding text notes Modify text notes Model text Editing model text **Tag** Tag tools Applying tag by category Tag all not tagged Material Tag Modifying tags **Keynote** Understanding keynote Types of Keynote Placing Keynotes Keynoting Settings Custom Keynotes Adding Keynote in 3D view **Callout Views** Understanding callout views Types of Callouts **Detailing** Creating detail views Creating drafting views Importing views from CAD program Reusing drafting view Inserting detail components Repeating detail	Theory/Demo	2.0 Hrs
21	Session 26	Exercises included for Practice	Practical	2.0 Hrs
		Schedules Creating Schedules/quantities Specifying Schedule properties		

22	Session 27	Custom parameters Modify schedules Material take off schedules Creating annotation schedule Reusing schedule views **Rooms** Creating rooms Room boundaries Room volume Situations that can effect Room volume computations Room tag Schedule Keys **Area** Area schemes Area types Area plans Area tags Applying color scheme Color fill legend Understanding Legends	Theory/Demo	2.0 Hrs
23	Session 28	Exercises included for Practice	Practical	2.0 Hrs
24	Session 29 & 30	**Sheets** Creating title blocks Revision workflow Revision schedules on title block Modifying views on sheet Rotating views on sheet Dividing view across multiple sheets Sheet List **Print** Print Setup Print Preview Printing Views and Sheets **Lights** Lighting fixtures Defining geometry of light source Controlling the brightness of light source Light groups **Materials** Understanding and Creating material library About the material properties and assets Editing material properties Editing assets Creating material Applying material to elements Changing the graphic properties of a material Changing the appearance properties of a material **Sun settings** Specifying sun settings Using sun path **Rendering** Defining area to render Specifying render quality Defining custom render quality Daylight portals Specifying output settings Lighting settings Background for render image Exporting rendered image to a file **Applying Decals** **Creating and Editing Walkthroughs** **Solar study** Setting up and creating solar study Rotating project north Displaying Sun and shadows	Theory/Demo	4.0Hrs
25	Session 31	Exercises included for Practice	Practical	2.0 Hrs
26	Session 32	Exercises included for Practice	Practical	2.0 Hrs

27	Session 33	**Site Design** Site Settings Topo surface Toposurface Subregions Splitting a toposurface Merging toposurface Building pads Property lines Contour line labels Parking components Graded region Site components **Working in a team** Work sharing over view Work sharing terminology Work sharing workflow Work sharing display modes **Working with linked models** Link Revit Visibility of linked models Manage links Shared positioning Project base point and survey points Working with point clouds files Import/ Link CAD Export to various formats	Theory/Demo	2.0 Hrs
28	Session 34	Exercises included for Practice	Practical	2.0 Hrs
29	Session 35	**Design options** Design options workflow Design option terminology Consideration when using design options Room areas and perimeter for design options Design options and wall joins **Phasing** Phasing workflow Phase properties Demolish elements Transferring project standards Interference Check **Customizing project settings** Fill patterns, Line weights, Line pattern, Line styles **Purge unused objects** Deleting unused Families and Family Types **Groups** Create groups, Modify groups, Saving groups, Loading groups	Theory/Demo	2.0 Hrs
30	Session 36	Exercises included for Practice	Practical	2.0 Hrs
31	Session 37	**Family creation** Reference planes Reference lines Constraints and dimensions **Creating door family** Drawing the door plan view components Creating door panel solid geometry Creating swing door family **Window family creation** Sliding window Creating arched window Window family **Creating new furniture family** Table family **Lighting fixture** Creating lighting fixture with one light source Creating lighting fixtures with multiple lightsources **Face based family** Kitchen cabinet Door handle **Custom staircase**	Theory/Demo	2.0 Hrs

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		Stair nosing, custom handrail, custom baluster		
32	Session 38	Exercises included for Practice	Practical	2.0 Hrs
33	Session 39	**Working with adaptive components** Adaptive components Curtain panel pattern 1 Curtain panel pattern 2.0 Curtain panel pattern 3 Creating Helix Creating Grill design Railing on curved surface Reporting parameter Dividing paths by distribution algorithm Component repetition Adaptive curved railing **Updates in Revit Architecture**	Theory/Demo	2.0 Hrs
34	Session 40	Exercises included for Practice	Practical	2.0 Hrs